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 <https://github.com/Arun-21BEECE006>

 <https://arun-portfolio-chi.vercel.app>



Summary

Software Development Engineer with experience building scalable full-stack applications using Python (Django, Flask) and modern frontend frameworks (React, Angular). Built and deployed DevLink — a production-ready enterprise workforce management platform replacing Jira, Slack and Zoom with real-time communication, AI-powered analytics and role-based access control. Currently working as Junior AI/ML Engineer at NGXP Technologies, developing real-time computer vision systems for L&T using YOLOv8. Experienced in RESTful APIs, WebSockets, WebRTC, AI/ML integration and enterprise web development with a focus on performance, security and reliability.

Experience

NGXP Technologies

(May 2026 – Present)

Junior AI/ML Engineer

Kochi, Kerala

- Developing a real-time camera-based vehicle and person detection system for L&T (Larsen & Toubro) using YOLOv8, enabling automated surveillance and activity monitoring across industrial sites
- Building and annotating custom training datasets using Roboflow, including image labelling, augmentation and dataset versioning for multi-class object detection (vehicles, persons, safety gear).
- Training and fine-tuning YOLO models on domain-specific datasets to achieve high accuracy detection under varied lighting and environmental conditions.
- Integrating the detection pipeline with live CCTV camera feeds for real-time inference, alert generation and event logging.

Mitsogo Technologies Pvt Ltd (Hexnode)

(July 2025 – March 2026)

Software Engineer

Kochi, Kerala

- Developed and maintained scalable full-stack features for the Hexnode Unified Endpoint Management (UEM) Portal using Django (Backend) and Angular (Frontend).
- Built and integrated secure RESTful APIs for device management, authentication, and portal operations, ensuring seamless communication between frontend and backend systems.
- Implemented core portal modules including user management, access control, and security enhancements, improving system reliability and performance.
- Collaborated in an Agile development environment with cross-functional teams to deliver production-ready features within strict deadlines and enterprise-grade standards.

Projects

DevLink - Enterprise Workforce Management Platform | Python, Flask, Flask - Socket IO, PostgreSQL, WebRTC, Gemini AI, JWT, React

(April 2026 – June 2026)

 **Live Demo:** <https://devlink.mo00.com>

- Built a production-deployed full-stack enterprise platform replacing Jira, Slack, Zoom and Asana with a single unified system, featuring role-based dashboards for Company Owner, Admin, Manager, Team Leader and Employee.
- Engineered real-time team communication using Flask-Socket IO including direct messaging, group chat, and peer-to-peer video calls via WebRTC with zero third-party service dependency.
- Integrated Google Gemini AI with a 2-key rotation system to power an intelligent assistant that answers team performance queries and auto-generates weekly reports.
- Implemented a full task lifecycle system with Kanban board, story points, sub-task delegation, approval workflows and performance tracking across all role levels.
- Deployed on Render with PostgreSQL on Neon.tech, FreeDNS custom domain and UptimeRobot keep-alive for 24/7 uptime

-  **Live Demo:** <https://ats-resume-analyzer-dp6l.onrender.com>
- Built a Flask-based ATS platform that extracts and analyzes resumes (PDF/DOCX) using PyPDF2 and python-docx for automated text processing.
 - Designed an NLP-driven scoring engine using keyword matching and regex to evaluate resume sections such as skills, experience, education, and projects.
 - Developed a dynamic job role recommendation system with weighted scoring, section-wise analysis, and personalized feedback for resume improvement.

Oral Cancer Detection System using CNN-Based Image Processing | MATLAB, LSTM, CNN, SqueezeNet, ResNet50, Image Processing

(December 2024 - February 2025)

-  **GitHub:** <https://github.com/Arun-21BEECE006/Oral-Cancer-using-LSTM-Full-Code>
- Developed a MATLAB-based oral cancer detection system using deep learning models (ResNet50 and SqueezeNet) for binary classification of medical images.
 - Applied image preprocessing techniques including resizing, grayscale conversion, Gaussian filtering, and normalization to enhance input X-ray/image quality.
 - Implemented CNN-based feature extraction with class activation mapping (CAM) and evaluated performance using sensitivity, specificity, and F-score metrics.

Skills	Achievements
• Frontend: HTML, CSS, JavaScript, React, Angular • Backend: Django, Flask, Node.js, Flask-Socket IO • Languages: C, C++, Java, Python • AI/ML: Yolo, NLP, TensorFlow, Gemini AI • Database: SQL, MongoDB, PostgreSQL, MySQL • Tools: Git, Postman, Eclipse, WebRTC, Roboflow	• Won First Place in All India Neo Codeathon – February Edition; awarded a smartwatch and certificate for outstanding performance. • Secured First Rank in II Semester and received Certificate of Appreciation. • Secured First Rank in III Semester and received Certificate of Appreciation

Internships

Motion Cut (AICTE Internship) <i>Web Development</i> • Developed responsive web applications using HTML, CSS, and JavaScript, focusing on user-friendly UI design and interactive components. • Built functional business websites and e-commerce interfaces with form validation and responsive layouts across device.	(November 2023 – December 2023)
Emglitz Technologies <i>Embedded C</i> • Developed embedded system applications using Embedded C, focusing on microcontroller-based programming and hardware interaction. • Implemented real-time embedded logic for sensor-based systems, strengthening hardware–software integration skills.	(May 2023 – June 2023) Coimbatore

Education

B.E Electronics and Communication Engineering Karpagam Academy of Higher Education, Coimbatore	(2021 – 2025) CGPA 9.43
HSC S.H.N.V Higher Secondary School, Sivakasi	(2020 – 2021) Percentage 87
SSLC S.H.N.V Higher Secondary School, Sivakasi	(2018 – 2019) Percentage 79